

WEB TOOL • OPEN SOURCE

Build causal models and reason about why.

The CAUSAL KNOWLEDGE BASE EDITOR is a visual interface to construct causal knowledge bases, **reason** over them through their induced argumentation framework, and get **human-readable explanations**.



SCAN TO TRY IT LIVE

causal-knowledge-base-editor.aig.fernuni-hagen.de

HOW IT WORKS

1 Causal model

Boolean-valued variables are tied together by *structural equations* and assumptions on uncontrollable background atoms.



2 Induced AF

Minimal, consistent arguments are derived from the model; undercuts between them yield an argumentation framework.



3 Reason & explain

Stable extensions determine which conclusions hold and support dependency and sequence-based explanations.

FEATURES

Model

Construct causal models graphically — atoms, structural equations, automatic layout.



Assume & observe

Fix assumptions on background atoms and choose the observations to reason from.



Reason

Valid conclusions computed from the stable extensions of the induced AF.



Explain

Dependency explanations and sequence-based explanations of conclusion arguments.



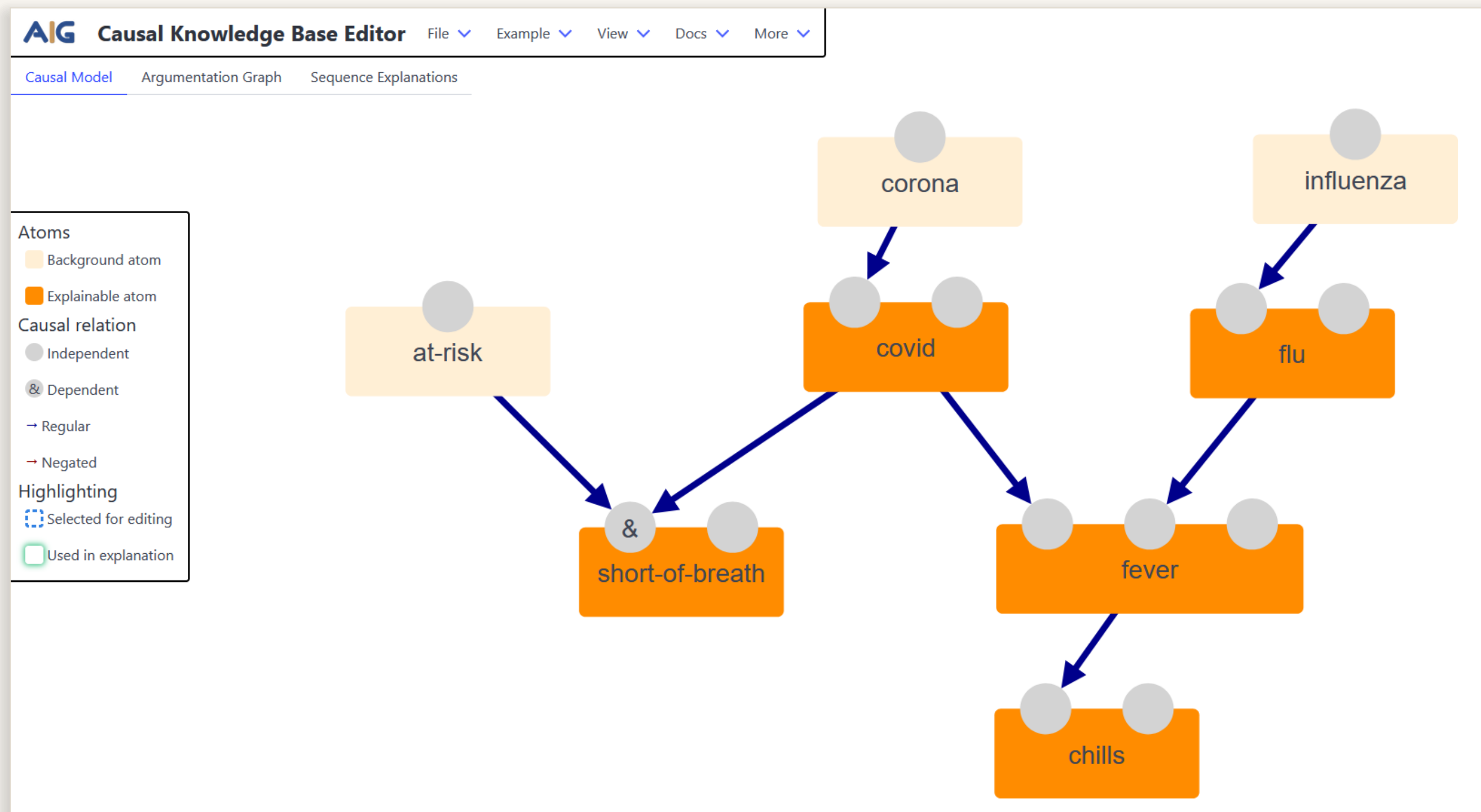
Switch views

Flip freely between the causal model and the induced AF behind it.



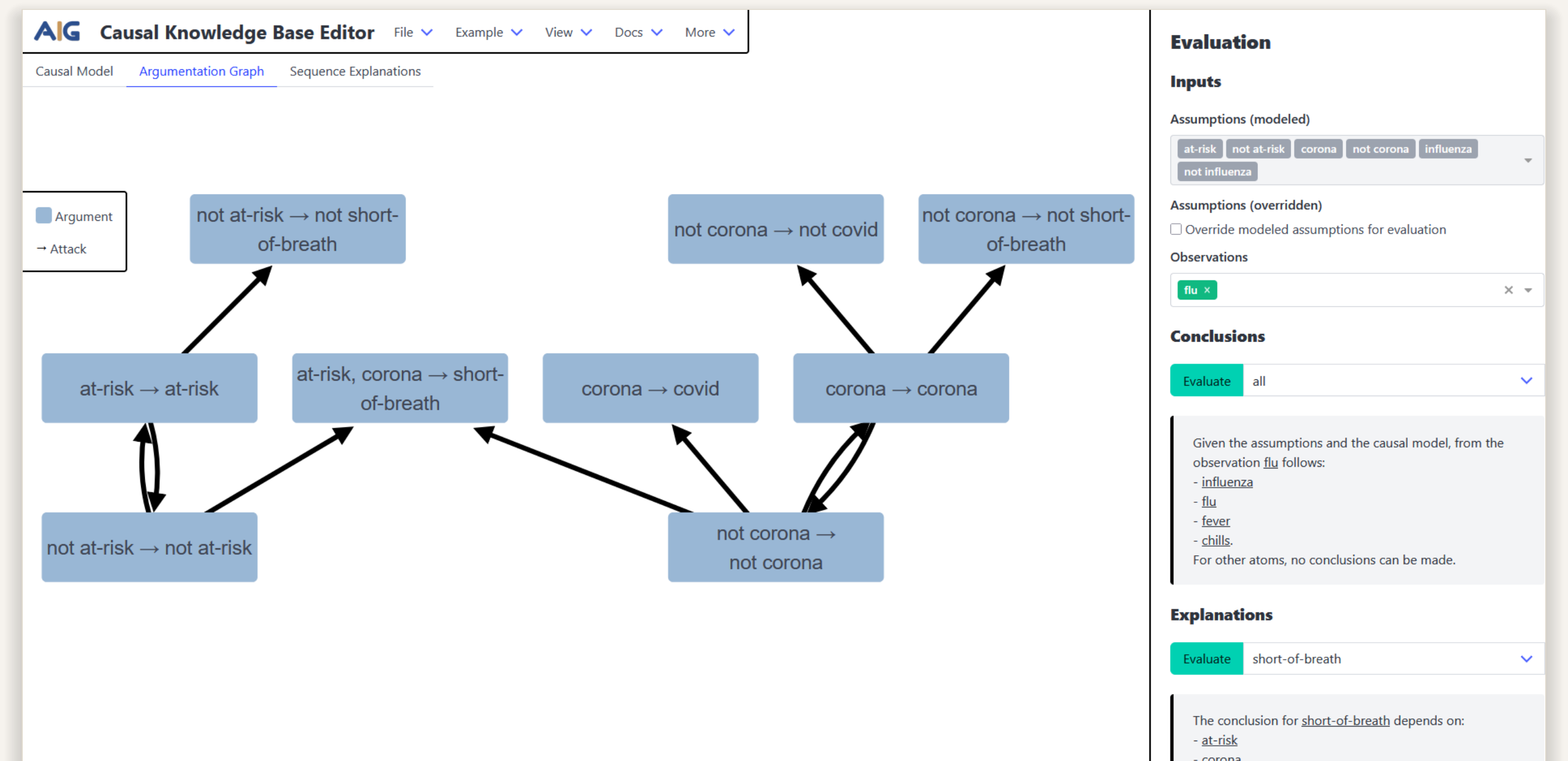
IN ACTION

1 Causal model



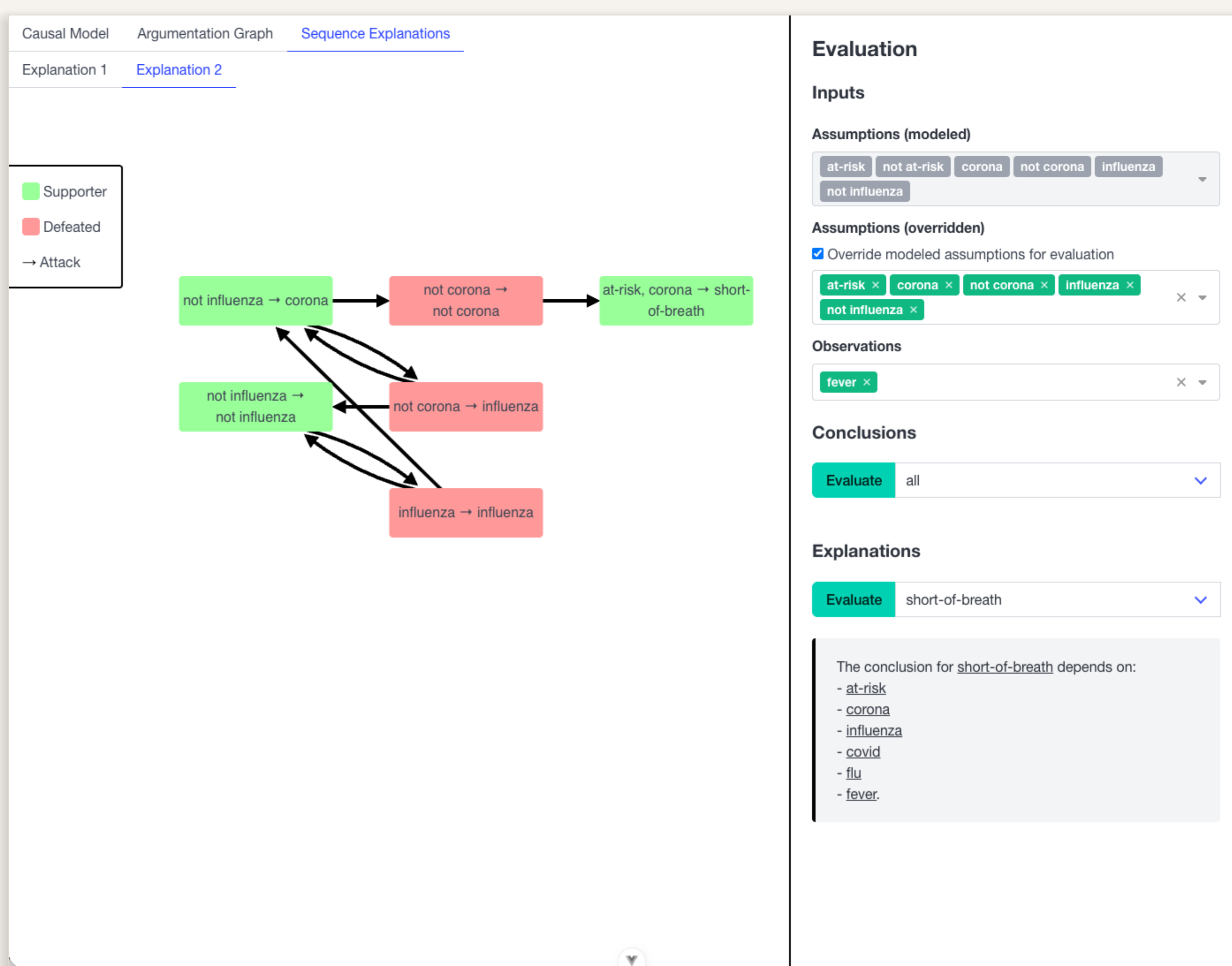
Build the structural equations and background assumptions graphically.

2 Induced AF



Inspect the arguments and attacks induced by the causal model.

3 Reason & explain



Evaluate conclusions and explore their sequence-based explanations.

OUTLOOK

- richer presentation and explanation of results within the AF
- counterfactual reasoning via the twin-model method

UNDER THE HOOD

A JavaScript front end built on the open-source `aig-graph-component`; causal reasoning and AF semantics by *TweetyProject*.

Open source — graphical modelling made to be explored, extended and reused.